

Experiment No. 5

Feature Extraction and Image Analysis using SIFT and HOG

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In [ ]: import cv2
import numpy as np
import matplotlib.pyplot as plt
from skimage.feature import hog
from skimage import exposure
# 1. Load Image
image_path = "image.jpg.png"
image = cv2.imread(image_path)
if image is None:
    print("Image not found! Please check the image path.")
else:
    print("Image loaded successfully!")
    # 2. Convert BGR to RGB
    rgb_image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    # Convert image to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    # Display original and grayscale image
    plt.figure(figsize=(12, 5))
    plt.subplot(1, 2, 1)
    plt.imshow(rgb_image)
    plt.title("Original Image")
    plt.axis("off")
    plt.subplot(1, 2, 2)
    plt.imshow(gray, cmap="gray")
    plt.title("Grayscale Image")
    plt.axis("off")
    plt.show()
    # 3. SIFT Feature Extraction
    sift = cv2.SIFT_create()
    # Detect keypoints and compute descriptors
    keypoints, descriptors = sift.detectAndCompute(gray, None)
    print("\nSIFT Feature Extraction")
    print("-----")
    print("Number of Keypoints:", len(keypoints))
    if descriptors is not None:
        print("Descriptor Shape:", descriptors.shape)
    # Draw SIFT keypoints
    sift_image = cv2.drawKeypoints(
        rgb_image,
        keypoints,
        None,
        flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS
    )
    # Display SIFT keypoints
    plt.figure(figsize=(8, 6))
    plt.imshow(sift_image)
    plt.title("SIFT Keypoints")
    plt.axis("off")
    plt.show()
    # 4. HOG Feature Extraction
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hog_features, hog_image = hog(
    gray,
    orientations=9,
    pixels_per_cell=(8, 8),
    cells_per_block=(2, 2),
    block_norm="L2-Hys",
    visualize=True
)
print("\nHOG Feature Extraction")
print("-----")
print("Number of HOG Features:", len(hog_features))
print("HOG Feature Shape:", hog_features.shape)
# Improve HOG visualization
hog_image_rescaled = exposure.rescale_intensity(
    hog_image,
    in_range=(0, 10)
)
# Display HOG image
plt.figure(figsize=(8, 6))
plt.imshow(hog_image_rescaled, cmap="gray")
plt.title("HOG Descriptor Visualization")
plt.axis("off")
plt.show()
# 5. Compare SIFT and HOG
print("\nComparison of SIFT and HOG")
print("-----")
print("SIFT Keypoints:", len(keypoints))
print("SIFT Descriptor Dimension:",
      descriptors.shape if descriptors is not None else "None")
print("HOG Feature Dimension:", len(hog_features))
print("\nSIFT:")
print("- Detects distinctive keypoints")
print("- Robust to scale and rotation changes")
print("- Useful for image matching and object recognition")
print("\nHOG:")
print("- Represents local gradient orientations")
print("- Captures shape and edge information")
print("- Useful for object detection and recognition")

```

Image loaded successfully!



SIFT Feature Extraction

 Number of Keypoints: 9437
 Descriptor Shape: (9437, 128)

SIFT Keypoints

Original Image

Low-Pass Filters (Smoothing & Noise Reduction)

- Gaussian Blur**: (Reduces noise, preserves structure)
- Median Filter**: (Removes salt-and-pepper noise, keeps edges)
- Average Filter**: (Single smoothing, can blur edges)

High-Pass Filters (Edge Enhancement & Feature Detection)

- Laplacian Filter**: (Enhances edges and fine details)
- Sobel X (Vertical Edges)**: (Detects vertical edges)
- Sobel Y (Horizontal Edges)**: (Detects horizontal edges)
- Edge Enhanced (Laplacian)**: (Sharper details and edges)

Salt-and-Pepper Noise Removal (Median Filter)

- Original**: (Salt-and-pepper noise)
- Filtered Image**: (Noise removed, edges preserved)

Filter	Type	Noise Reduction	Edge Preservation	Edge Enhancement	Main Application
Gaussian	Low-pass	Good	Good	Low	Smoothing, denoising
Median	Low-pass	Good	Good	Low	Removal of salt-and-pepper noise
Average	Low-pass	Good	Good	Low	Smoothing, denoising
Laplacian	High-pass	Low	Good	Good	Edge detection, feature extraction
Sobel X	High-pass	Low	Good	Good	Vertical edge detection
Sobel Y	High-pass	Low	Good	Good	Horizontal edge detection

Key Observations

- Low-pass filters (Gaussian, Median, Average) are effective for smoothing and noise reduction but may blur edges.
- High-pass filters (Laplacian, Sobel X, Sobel Y) are effective for edge detection and feature extraction but may lose some texture information.
- The Laplacian filter is particularly sensitive to noise and may produce artifacts.
- The Sobel filters are robust to noise and provide good edge detection results.
- The Edge Enhanced filter (Laplacian) provides a good balance between edge detection and noise reduction.
- The Salt-and-Pepper Noise Removal filter (Median Filter) is effective for removing salt-and-pepper noise while preserving edges.

HOG Feature Extraction

Number of HOG Features: 873252

HOG Feature Shape: (873252,)

HOG Descriptor Visualization



Comparison of SIFT and HOG

SIFT Keypoints: 9437

SIFT Descriptor Dimension: (9437, 128)

HOG Feature Dimension: 873252

SIFT:

- Detects distinctive keypoints
- Robust to scale and rotation changes
- Useful for image matching and object recognition

HOG:

- Represents local gradient orientations
- Captures shape and edge information
- Useful for object detection and recognition

Questions and Answers

1. What is feature extraction, and why is it important in computer vision?

Feature extraction is the process of identifying important information or characteristics from an image, such as edges, corners, shapes, textures, and patterns.

It is important because it reduces unnecessary image information and provides useful features that can be used for image matching, classification, object detection, and recognition.

2. Explain the working principle of Scale-Invariant Feature Transform (SIFT).

SIFT is a feature extraction technique used to detect and describe important points in an image. Its main steps are:

1. **Scale-Space Construction:** The image is analyzed at different scales using Gaussian blurring.
2. **Keypoint Detection:** Important points are detected using the Difference of Gaussian (DoG).
3. **Keypoint Localization:** Unstable or low-contrast keypoints are removed.
4. **Orientation Assignment:** An orientation is assigned to each keypoint based on local gradients.
5. **Feature Description:** A descriptor is created around each keypoint to represent its local image information.

SIFT produces features that are robust to changes in scale, rotation, and moderate changes in illumination.

3. What are keypoints and feature descriptors in image analysis?

Keypoints are important locations in an image, such as corners, edges, or regions with distinctive patterns. They can be detected using methods such as SIFT.

Feature descriptors are numerical representations of the area surrounding a keypoint. They describe the local characteristics of that region and can be used to compare features between different images.

4. Explain the concept of Histogram of Oriented Gradients (HOG) and its significance.

HOG is a feature descriptor that represents an image using the direction and magnitude of local gradients.

The image is divided into small cells, and the gradient directions of pixels are calculated. These directions are grouped into a histogram. Histograms from neighboring cells are normalized to improve robustness.

HOG is significant because it captures the shape and edge information of objects and is particularly useful for detecting objects such as pedestrians.

5. Compare SIFT and HOG based on robustness, computational complexity, and practical applications.

SIFT	HOG
Detects distinctive keypoints and creates descriptors.	Describes gradient and edge information over image regions.
Robust to scale and rotation changes.	Less robust to large scale and rotation changes.
Generally more computationally complex.	Generally simpler and faster.
Suitable for image matching and object recognition.	Commonly used for object and pedestrian detection.
Works well with local features.	Works well with shape and edge features.

6. Why is SIFT considered invariant to scale and rotation?

SIFT is scale-invariant because it searches for keypoints across multiple image scales using a scale-space representation.

It is rotation-invariant because an orientation is assigned to each keypoint based on the local gradient directions. The descriptor is then generated relative to this orientation, allowing the same feature to be recognized even when the image is rotated.

7. Mention three real-world applications where HOG descriptors are commonly used.

Three common applications of HOG are:

1. **Pedestrian Detection:** Detecting people in surveillance and traffic systems.
2. **Vehicle Detection:** Identifying vehicles based on their shape and edge patterns.
3. **Object Recognition:** Detecting objects with distinctive shapes and structures.

8. Why is feature extraction performed before image classification or object detection?

Raw images contain a large amount of information, including unnecessary details and noise. Feature extraction identifies the most important characteristics of an image.

These features provide a smaller and more meaningful representation that makes classification and object detection more efficient and accurate.

9. What are the advantages and limitations of handcrafted feature descriptors compared to deep learning-based feature extraction?

Advantages of Handcrafted Features:

- Require less training data.
- Easier to understand and implement.
- Can work well for specific applications.
- Usually require fewer computational resources.

Limitations:

- Depend on manually designed rules.
- May not perform well on complex images.
- Less adaptable to different types of data.
- Often perform worse than deep learning methods on large and diverse datasets.

Deep learning methods automatically learn useful features from training data and generally provide better performance for complex computer vision tasks.

10. How do feature extraction techniques contribute to image matching, face recognition, and object detection systems?

Feature extraction converts images into meaningful numerical representations.

- **Image Matching:** Features such as SIFT descriptors can be compared to find similar regions between images.
- **Face Recognition:** Features representing facial shapes and patterns help distinguish different faces.
- **Object Detection:** Features such as HOG describe object shapes and edges, helping detection algorithms identify objects.

Thus, feature extraction provides important information that improves the accuracy and efficiency of computer vision systems.